

Reg No.: _____

Name: _____

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APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Fourth Semester B.Tech Degree Examination July 2021 (2019 Scheme)

Course Code: MAT206

Course Name: GRAPH THEORY

Max. Marks: 100

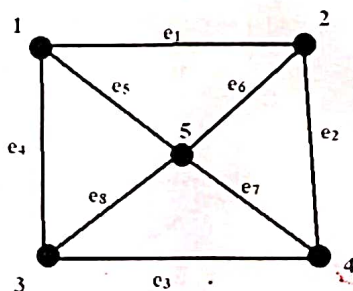
Duration: 3 Hours

PART A

(Answer all questions; each question carries 3 marks)

Marks

- | | | |
|---|--|---|
| 1 | What is the maximum number of edges in a simple graph with n vertices? Justify your answer. | 3 |
| 2 | There are 25 telephones in Metropolis. Is it possible to connect them with wires so that each telephone is connected with exactly 7 others? Why? | 3 |
| 3 | Show that all vertices of an Euler graph G are of even degree | 3 |
| 4 | Explain strongly connected and weakly connected graphs with the help of examples. | 3 |
| 5 | Prove that a connected graph G with n vertices and $n-1$ edges is a tree. | 3 |
| 6 | How many labelled trees are there with n vertices? Draw all labelled trees with 3 vertices. | 3 |
| 7 | Define planar graphs. Is K_4 , the complete graph with 4 vertices, a planar graph? Justify. | 3 |
| 8 | Define fundamental circuits and fundamental cut-sets. | 3 |
| 9 | Construct the adjacency matrix and incidence matrix of the graph . | 3 |



- | | | |
|----|--|---|
| 10 | Define chromatic number. What is the chromatic number of a tree with two or more vertices? | 3 |
|----|--|---|

PART B

(Answer one full question from each module, each question carries 14 marks)

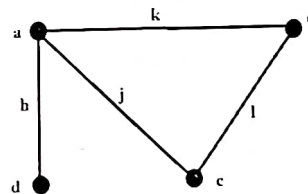
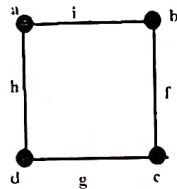
Module -1

defn ②+③ + eg 1-2(2)

- 11 a) Define complete graph and complete bipartite graph. Draw a graph which is a complete graph as well as a complete bipartite graph. 7
- b) Explain walks, paths and circuits with the help of examples. 7
- 12 a) Define isolated vertex, pendant vertex, even vertex and odd vertex. Draw a graph that contains all the above. 7
- b) Prove that simple graph with n vertices and k components can have at most $(n-k)(n-k+1)/2$ edges. 7

Module -2

13 a)

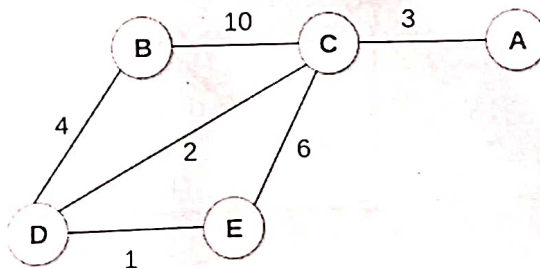


Find the union, intersection and ring sum of the above graphs.

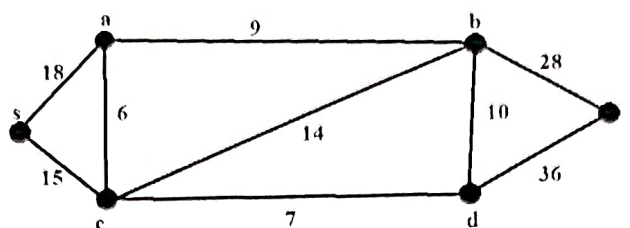
- b) State travelling salesman problem. How it is related to Hamiltonian circuits? 5
- 14 a) Prove that in a complete graph with n vertices there are $(n-1)/2$ edge disjoint Hamiltonian circuits, if n is an odd number and $n \geq 3$. 7
- b) For which values of m, n is the complete graph $K_{m,n}$ an Euler graph? Justify your answer. 7

Module -3

- 15 a) Prove that a binary tree with n vertices has $(n+1)/2$ pendant vertices. 7
- b) Using Prim's algorithm, find a minimal spanning tree for the following graph. 7



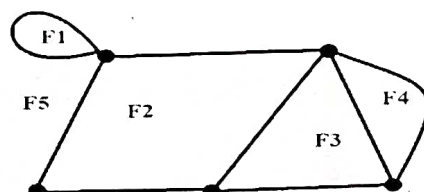
- 16 a) Write down Dijkstra's algorithm and use it to find the shortest path from s to t . 9



- b) Prove that every tree has either one or two centers. 5

Module -4

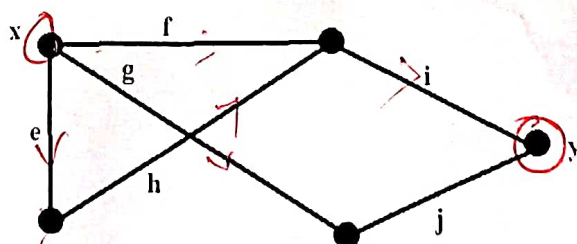
- 17 a) Define cut-set. Prove that every circuit in G has an even number of edges in common with any cut-set. 8
- b) Construct the geometric dual of the graph below 6



- 18 a) Prove that a connected planar graph with n vertices and e edges has $e-n+2$ regions. 9
- b) Let G be a connected graph and e an edge of G . Show that e is a cut-edge if and only if e belongs to every spanning tree. 5

Module -5

- 19 a) Explain *four colour problem* using the concept of chromatic number. 5
- b) Let B and A be the circuit matrix and the incidence matrix of a graph G which is free from loops, whose columns are arranged using the same order of edges. Show that $AB^T = BA^T = 0 \pmod{2}$. 9
- 20 a) Show that chromatic polynomial of a tree with n vertices is $P_n(\lambda) = \lambda(\lambda - 1)^{n-1}$ 7
- b) Define path matrix of a graph. Find the path matrix $P(x, y)$ for the graph below. 7



$$\begin{matrix} & e & f & g & h & i & j \\ \begin{matrix} 1 \\ 2 \\ 3 \end{matrix} & \begin{bmatrix} 0 & 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 0 & 0 & 1 \end{bmatrix} \end{matrix}$$

$\{f, i\}$
 $\{e, h, i\}$
 $\{g, j\}$



APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Scheme for Valuation/Answer Key

Scheme of evaluation (marks in brackets) and answers of problems/key

FOURTH SEMESTER B.TECH DEGREE EXAMINATION JULY 2021 (2019 Scheme)

Course Code: MAT206

Course Name: GRAPH THEORY

Max. Marks: 100

Duration: 3 Hours

PART A

(Answer all questions; each question carries 3 marks)

Marks

- | | | |
|---|---|---|
| 1 | Max number of edges is $n(n-1)/2$ -----(1)
Justification------(2) | 3 |
| 2 | Not possible------(1)
No of edges in such a graph is $(25*7)/2$ which is not an integer -----
(2) | 3 |
| 3 | G is Euler $\Rightarrow$ G contains a Euler line------(1)
Argue that degree of internal vertex of the trail is a multiple of two -----
--(1)
The starting and ending vertex of the Euler line is the same -----
---(1) | 3 |
| 4 | Strongly connected-a directed path from u to v for any pair of vertices (u,v) of G-(1)
Weakly connected-the underlying graph is connected-----
(1)
Examples for both------(1) | 3 |
| 5 | Assume that G is not a tree $\Rightarrow$ G has cycles------(1)
$\rightarrow$ delete an edge from each cycles one by one so that the graph remains connected------(1)
$\rightarrow$ We have a connected acyclic graph (ie a tree) with fewer edges than $(n-1)$, a contradiction.------(1) | 3 |
| 6 | There are n^{n-2} labelled trees in n vertices------(2)
Drawing all 3 labelled trees------(1) | 3 |
| 7 | G is planar if there exist a geometric representation which can be drawn on a plane in such a way that no edges intersects.----- | 3 |



- (1)
 K_4 is a planar graph------(1)
 Drawing the planar embedding------(1)
 Definition of fundamental circuit -----(1) 3
 Definition of fundamental cut set------(2)
 Adjacency matrix------(1.5) 3

$$\begin{bmatrix} 0 & 1 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 & 1 \\ 0 & 1 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 & 0 \end{bmatrix}$$

Incidence matrix-----

$$\begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \end{bmatrix} \text{------(1.5)}$$

- 10 The minimum number of colours required to properly colour a graph is 3
 known as chromatic number-----
 ----(1)
 The chromatic number of a tree with two or more vertices is $\kappa=2$ ------(2)

PART B

(Answer one full question from each module, each question carries 14 marks)

Module -1

- 11 a) Definition -----complete graph------(2) 7
 Complete bipartite graph ------(3)
 Giving K_2 as the example ------(2)
 b) Definitions of walks, paths and circuits------(4) 7
 Examples for each------(3)
 12 a) Definition of isolated, pendant, even, odd vertices.....(4) 7
 Drawing a graph containing all the above------(3)
 b) Let each component contains $n_1, n_2, \dots, n_k$ vertices. 7
 Max edges in i^{th} component is $n_i(n_i-1)/2$ ------(4)
 Proving $\sum_{i=1}^k n_i(n_i - 1)/2 \leq (n-k)(n-k+1)/2$ ------(3)



Module -2

13 a)



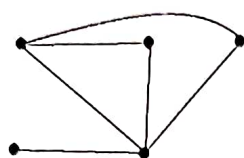
------(3)



9



-----(3)



-(3)

5

b) Stating travelling salesman problem------(2)

Formulating the problem using Hamiltonian circuits------(3)

14 a) Number of edge disjoint circuits cannot exceed $(n-1)/2$ ------(2) 7Keep the vertices in a circle, rotate clockwise $2\pi/(n-1)$ each time to produce the edge disjoint Hamiltonian circuits-----

------(3)

Conclusion -----(2)

b) Graph is Euler iff degree of every vertex is even.----- 7

--(2)

Degree of a vertex in $K_{m,n}$ is either m or n.-----

--(3)

 $K_{m,n}$ is Euler iff m & n are even.-----

-(2)

Module -3

15 a) p-no of pendant vertices; n-p-1 --no of vertices with degree 3; one vertex 7

with degree 2,-----

(2)

Equating $[p+3(n-p-1)+2]=2(n-1)$ -----(3)

Solving for p -----(2)

b) Writing the Prim's algorithm ----- 7

-----(2)

Using prims algorithm to find minimal spanning tree-----



----- (2)

Giving the spanning tree-----

----- (2)

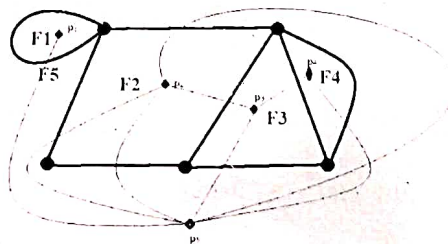
Weight of minimal spanning tree=10.-----

----- (1)

- 16 a) Writing down the algorithm -----(4) 9
Using Dijkstra's algorithm -----(3)
Shortest path $s \rightarrow a \rightarrow b \rightarrow t$ with length 55.-----(2)
- b) Deleting pendant vertices in the tree step by step to have a centre having one or two vertices. 5

Module -4

- 17 a) Definition of a cut set------(2) 8
- S be a cut set of G. Deletion of S partitioned the vertices of G into two sets V_1 & V_2 . Let C be a circuit in G. If all vertices of C is in V_1 (or V_2), then zero common edge with S-----
------(4)
- If not each travel of C from V_1 to V_2 and vice versa use a new edge from S.
- Since C starts and end in the same partition, the proof follows-----
-(2)



- 18 a) A connected planar graph has a representation in which each region is a polygon
- Let k_p represent p sided polygonal regions. Since each side is a boundary of two regions
- $$3 \cdot k_3 + 4 \cdot k_4 + \dots + r \cdot k_r = 2e \dots \dots \dots (2)$$
- Sum of all angles suspended at each vertex is $2\pi n$.



Also $\sum \pi(p-2)k_p + 4\pi = 2n\pi$ (with proper arguments)-----

(4)

Equating and solving $\pi(2e-2f)+4\pi=2n\pi$, we have the result.....(3)

Or

Any other correct proof

- b) Let G be a connected graph. 'e' be a cut edge of G ie $G-\{e\}$ is disconnected. 5

If e does not belong to a spanning tree of G , then $G-\{e\}$ contains that tree making $G-\{e\}$ connected, a contradiction.-----

(3)

Let e is in every spanning tree of G . If e is not a cut edge, then $G-\{e\}$ is connected and thus it contains a spanning tree of $G-\{e\}$, which do not contain e and also a spanning tree of G . a contradiction.-----

------(2)

Module -5

- 19 a) Stating four colour problem------(2) 5
Stating in terms of chromatic number------(3)

- b) Consider an arbitrary vertex v and arbitrary circuit C . If v is not in C , then 9
no edge of the circuit is incident on v and if v is in C then two edges of C are incident to v .------(2)

Consider i^{th} row in A and j^{th} row in B . The non zero entries in the corresponding positions occur only if the particular edge say e_r is incident on the i^{th} vertex and belongs to the j^{th} circuit.-----

------(2)

i. e_r is incident on v_i and $e_r \notin C_j$. Here $a_{ir} = 1$, $b_{jr} = 0$.

ii. e_r is not incident on v_i and $e_r \in C_j$. In this case, $a_{ir} = 0$, $b_{jr} = 1$.

iii. e_r is not incident on v_i and $e_r \notin C_j$, so that $a_{ir} = 0$, $b_{jr} = 0$.

All these cases imply that the i^{th} vertex v_i is not in the j^{th} cycle Z_j and we have

$$[AB^T]_{ij} = 0 \equiv 0 \pmod{2}.$$



iv. e_i is incident on v_i and $e_i \in C_j$. Here we have exactly two edges, say e_i and e_t incident on v_i so that $a_{it} = 1$, $a_{ti} = 1$, $b_{it} = 1$, $b_{ti} = 1$. Therefore, $[AB^T]_{ij} = 1+1 \equiv 0 \pmod{2}$

 -----(5)

- 20 a) Begin by colouring pendent vertex of the tree. There are λ choices of colour for the first vertex, and $(\lambda - 1)$ choices of colour for the second vertex which is adjacent to the first because it is adjacent to the first vertex. -----(2)

For the third vertex, we also have $(\lambda - 1)$ choices of colour because we cannot use the 1 colour which we used for the second vertex (to which the third vertex is adjacent).-----

------(4)

Similarly, we will have $(\lambda - 1)$ choices of colour for all of the remaining vertices

Hence the result------(1)

[We can also prove the result by induction on number of vertices of the tree]

- b) Defining path matrix $P(xy)=(p_{ij})$ where $p_{ij}=1$ if j^{th} edge belongs to the i^{th} path.-(2)

Identifying three paths in the graph------(2)

Writing down the path matrix------(3)

Note: Credit should be given to any valid proofs/solutions other than given in the key. Partial credits should be given to any valid attempt in the right direction.